Analysis of Protein–DNA Target Search and Polymer Dynamics

Internship Report

Submitted by

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Abstract

This report details the work done during my summer internship, focusing on analyzing trajectory data from molecular dynamics (MD) simulations of biophysical and polymer systems. The datasets were generated using LAMMPS, and I wrote custom Python scripts to study two main problems: how proteins search for target sites on DNA, and how polymer chains contract and rearrange during an annealing process.

The report covers four key tasks. First, I analyzed a polymer chain under equilibrium conditions to calculate baseline values for its Radius of Gyration (R_g), Persistence Length ($l_p = 0.98\sigma$), and Radial Distribution Function ($g(r)$). Second, I studied a protein–DNA target search model that uses facilitated diffusion. By running a parameter sweep across four interaction strengths ($\epsilon = 3, 4, 5, 6$) over 300 random seeds, I measured the average 1D sliding times, 3D bulk times, and switching frequencies. Third, I looked at non-equilibrium polymer relaxation across seven cooling blocks during thermal annealing, tracking how the chain shrinks and fitting Mean Squared Displacement (MSD) curves to find the subdiffusive scaling exponents. Fourth, I calculated the static structure factor ($S(\mathbf{k})$) to observe the scaling behavior ($\alpha \approx 1.3 - 1.6$) in reciprocal space across different length scales.

Because the raw LAMMPS trajectory files were quite large, I optimized the Python code using NumPy vectorization, multi-threading, and Fast Fourier Transforms (FFT) for the correlation functions to speed up processing times.

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Chapter 1

Introduction

1.1 Overview

Understanding how large biomolecules move and change their shape is important for a lot of problems in soft matter physics and biology. For example, we want to know how a protein quickly finds a small target sequence on a long, crumpled piece of DNA, or how a synthetic polymer chain packs together when it cools down. Since it is very hard to watch these processes directly in a lab, we use Molecular Dynamics (MD) simulations to look at individual particle positions over time.

During my internship at IIT Palakkad, my main job was to write and test Python pipelines to extract physical properties from raw simulation data. The datasets came from LAMMPS simulations and consisted of text files containing the 3D coordinates of all atoms across multiple timesteps. The work was split into two separate projects:

1. **Protein–DNA Target Search:** Tracking a mobile protein particle as it switches between 3D bulk diffusion and 1D sliding along a 101-monomer DNA strand to find its target.
2. **Polymer Annealing:** Analyzing how a dense polymer chain contracts and relaxes when it is systematically cooled down.

I used standard formulas from statistical mechanics to turn these massive coordinate files into meaningful plots and numbers. This report outlines the methods I used, the numbers I got out of the trajectories, the code optimizations I wrote, and how I verified that my scripts were actually correct. The overall workflow of the analysis pipeline is shown in Figure 1.1.

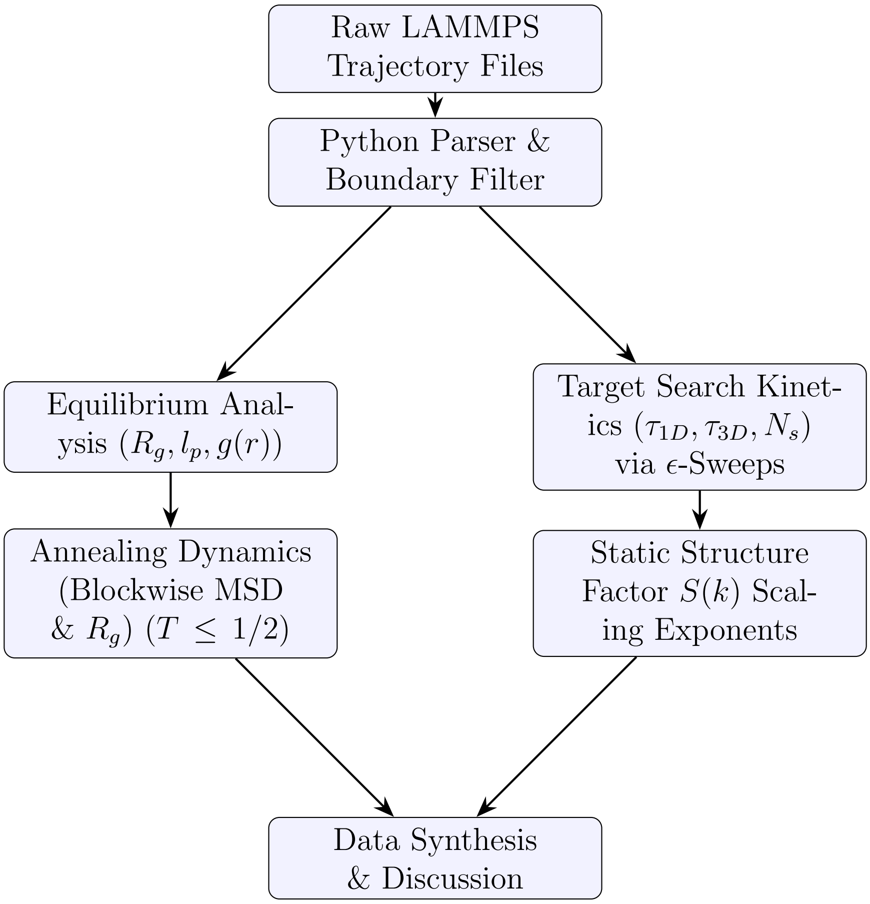

Figure 1.1: Flowchart of the trajectory parser and data analysis pipeline.

1.2 Simulation Models and Parameters

The datasets came from two separate coarse-grained simulation setups. Both configurations use reduced Lennard-Jones units, which means that physical values like mass, length (σ), and energy are dimensionless scalars.

1.2.1 System I: Protein–DNA Target Search

This system models how a protein finds its target sequence using implicit-solvent Brownian dynamics. The setup consists of a fixed, semi-flexible DNA strand containing 101 coarse-grained spherical monomers and a single mobile protein bead. The non-bonded interaction between the protein and the DNA is modeled with an attractive potential well where the depth, ϵ, can be adjusted. To see how the balance between 3D bulk diffusion and 1D sliding changes, four different interaction strengths ($\epsilon = 3, 4, 5, 6$) were simulated, using 300 independent random seeds for each setting. The main parameters for this setup are listed in Table 1.1.

Parameter	**Value**
Simulation Type	Coarse-Grained Brownian Dynamics
Software Used	LAMMPS
DNA Chain Length	101 monomers
Protein Count	1 bead
Total Particles (N)	102
Interaction Strengths (ϵ)	3, 4, 5, 6
Integration Timestep (dt)	$0.005\,\tau$
Dump Frequency	Every 1,000 steps
Time Between Frames (Δt)	$5.0\,\tau$
Boundary Conditions	Periodic (*pp pp pp*)
Simulation Box Size ($L_x \times L_y \times L_z$)	$50 \times 50 \times 150$
Coordinate Format	Cartesian

Table 1.1: Simulation settings for System I (Protein–DNA Target Search).

1.2.2 System II: Polymer Equilibrium and Annealing

This dataset was used to study how a polymer chain contracts and relaxes in a dense environment. The simulation box contained 2,100 total particles split into three types. Type 1 is the main polymer backbone (100 monomers), while Type 2 and Type 3 are surrounding crowder beads (1,000 particles each) added to simulate a crowded matrix.

The coordinates were recorded in unwrapped formats (`xu, yu, zu`) so that we could track the distance a particle traveled continuously without artificial jumps when crossing periodic box edges. The temperature was controlled using a Langevin thermostat. The production run went on for 1.75×10^8 total steps, saving 7,001 frames divided evenly across seven distinct cooling windows. Table 1.2 summarizes this setup.

Parameter	Value
Simulation Type	Coarse-Grained Molecular Dynamics
Software Used	LAMMPS
Trajectory File Name	`Nb100Ns20gd1.0.lmptrj`
Coordinate Format	Unwrapped (`xu, yu, zu`)
Total Particles (N)	2,100
Composition	Type 1 (100); Type 2 (1,000); Type 3 (1,000)
Starting Step	0
Ending Step	1.75×10^8
Total Saved Frames	7,001
Dump Frequency	Every 25,000 steps
Boundary Conditions	Periodic (*pp pp pp*)
Simulation Box Size ($L_x \times L_y \times L_z$)	$60 \times 40 \times 150$
Box Coordinates Range	$x \in [-30, 30],\ y \in [-20, 20],\ z \in [-75, 75]$
Thermostat	Langevin

Table 1.2: Simulation settings for System II (Polymer Annealing).

Chapter 2

Polymer Equilibrium Structural Analysis

2.1 Introduction and Objectives

My first task was to look at a baseline polymer chain under standard equilibrium conditions. The goals were to figure out the average physical size of the chain, measure its structural stiffness, and see how the monomers arrange themselves locally.

2.2 Simulation Data

I used the initial equilibrium trajectory files from LAMMPS, which listed the 3D coordinates of the monomers and the changing box boundaries. To keep the processing fast, I added a frame-skipping filter into the file parsing scripts so we didn't have to look at redundant frames.

2.3 Methodology

2.3.1 Radius of Gyration (R_g)

The Radius of Gyration shows how far the monomers are spread out from the chain's center of mass. For a chain of N identical atoms, R_g is calculated as:

$$R_g = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\mathbf{r}_i - \mathbf{r}_{cm})^2} \tag{2.1}$$

where $\mathbf{r}_i$ is the position of atom i, and $\mathbf{r}_{cm}$ is the center of mass of the chain at that instant:

$$\mathbf{r}_{cm} = \frac{1}{N} \sum_{j=1}^{N} \mathbf{r}_j \tag{2.2}$$

I used a simple moving average to smooth out the high-frequency thermal noise in the raw output.

2.3.2 Persistence Length Analysis

To measure the mechanical stiffness of the polymer chain, I looked at the bond-orientational correlation function. Let $\mathbf{t}_i$ be the normalized vector pointing along bond i from monomer i to $i+1$:

$$\mathbf{t}_i = \frac{\mathbf{r}_{i+1} - \mathbf{r}_i}{|\mathbf{r}_{i+1} - \mathbf{r}_i|} \tag{2.3}$$

The alignment between two bond vectors separated by a distance s along the chain backbone is expected to decay exponentially:

$$\langle \mathbf{t}_i \cdot \mathbf{t}_{i+s} \rangle \sim e^{-s/l_p} \tag{2.4}$$

where l_p is the persistence length. I extracted l_p by fitting an exponential curve to the short-range data points, leaving out the noisy long-range correlations where the signal degrades.

2.3.3 Radial Distribution Function (RDF)

The Radial Distribution Function, $g(r)$, measures local monomer packing and coordination shells. It is calculated by counting pairs inside a thin shell and dividing by the average density:

$$g(r) = \frac{N_{\text{pairs}}(r)}{\left(\frac{N(N-1)}{2V}\right) \Delta V(r)} \tag{2.5}$$

where $N_{\text{pairs}}(r)$ is the number of monomer pairs separated by a distance between r and $r + \Delta r$, V is the box volume, and $\Delta V(r) = \frac{4}{3}\pi \left[(r+\Delta r)^3 - r^3\right]$ is the shell volume. The code uses the minimum image convention to handle the periodic boundaries properly.

2.4 Results and Validation

The calculated numbers for the equilibrium run are summarized in Table 2.1.

Metric	Extracted Value	Physical Meaning
Initial Radius of Gyration (R_g)	$15.20\,\sigma$	Size at start of tracking window
Final Radius of Gyration (R_g)	$7.02\,\sigma$	Size after contraction settles
Persistence Length (l_p)	$0.98\,\sigma$	High flexibility; matches ideal chain expectatio
First RDF Peak Position (r)	$1.00\,\sigma$	Enforced covalent bond distance

Table 2.1: Quantitative summary of equilibrium polymer structural properties.

The polymer chain starts off in an expanded shape, with R_g sitting around 15.20σ in the early frames. As the trajectory progresses, R_g drops steadily down to about 7.02σ, which shows that the chain is compressing over time (Figure 2.1). The curve eventually flattens out into a horizontal line, meaning the structural contraction has slowed down and stabilized.

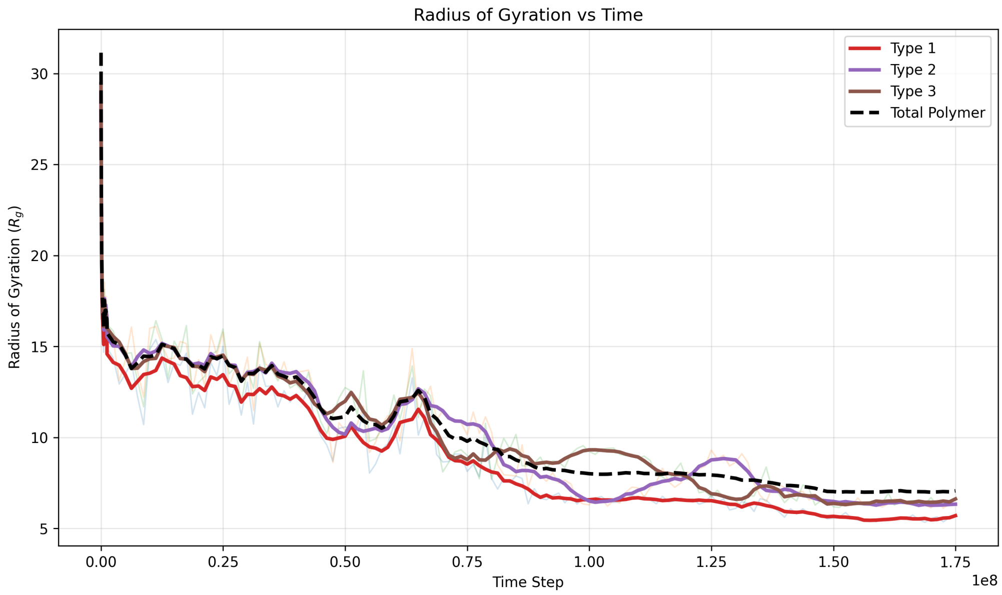

Figure 2.1: Radius of Gyration (R_g) vs Time showing the chain shrinking over the simulation run.

The bond correlation function falls off very fast as the separation distance s increases (Figure 2.2). The exponential fit gives a persistence length of $l_p = 0.98\sigma$.

Code Validation Note: To verify that my analysis script was working correctly, I used an ideal freely-jointed chain as a theoretical benchmark. Because the simulation topology lacks any explicit angle-bending potentials, the physical persistence length should equal the nominal bond distance of 1.00σ. The script returned 0.98σ, which is well within expected numerical error limits and confirms that the vector extraction and regression mathematics are correct.

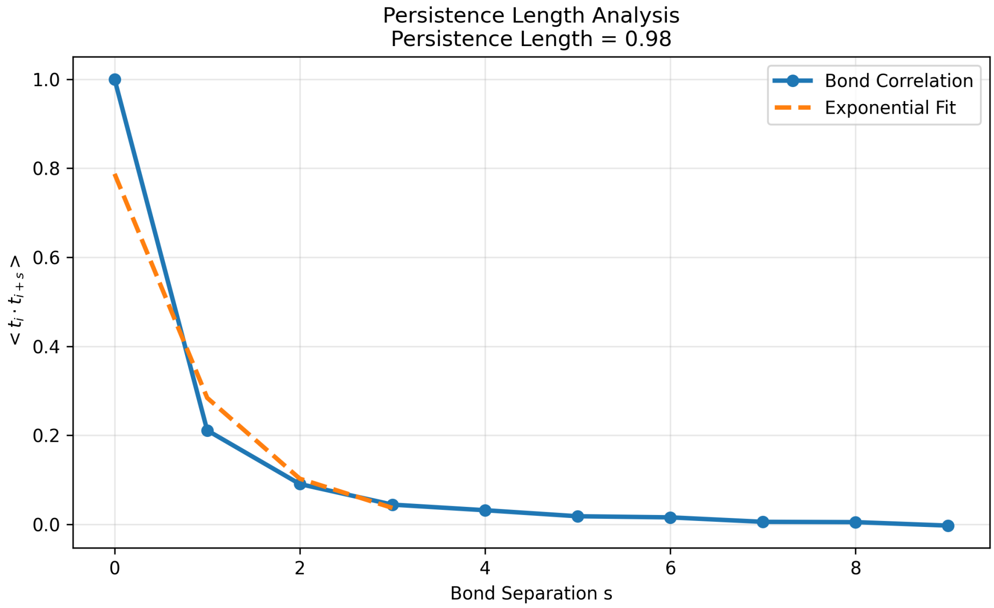

Figure 2.2: Exponential decay of the bond orientational correlation function yielding $l_p = 0.98$.

The $g(r)$ profile shown in Figure 2.3 features a very sharp, intense first peak exactly at $r = 1.00\sigma$. This matches the physical bond distance between adjacent backbone monomers (i and $i \pm 1$) perfectly, acting as a direct internal check on the distance calculation logic. The smaller peaks that follow represent the local coordination shells of non-bonded monomers packing around each other due to volume exclusion. At larger distances ($r > 3$), the oscillations flatten out to 1.0, which confirms that the system has a disordered, fluid-like arrangement at large distances instead of a rigid crystal structure.

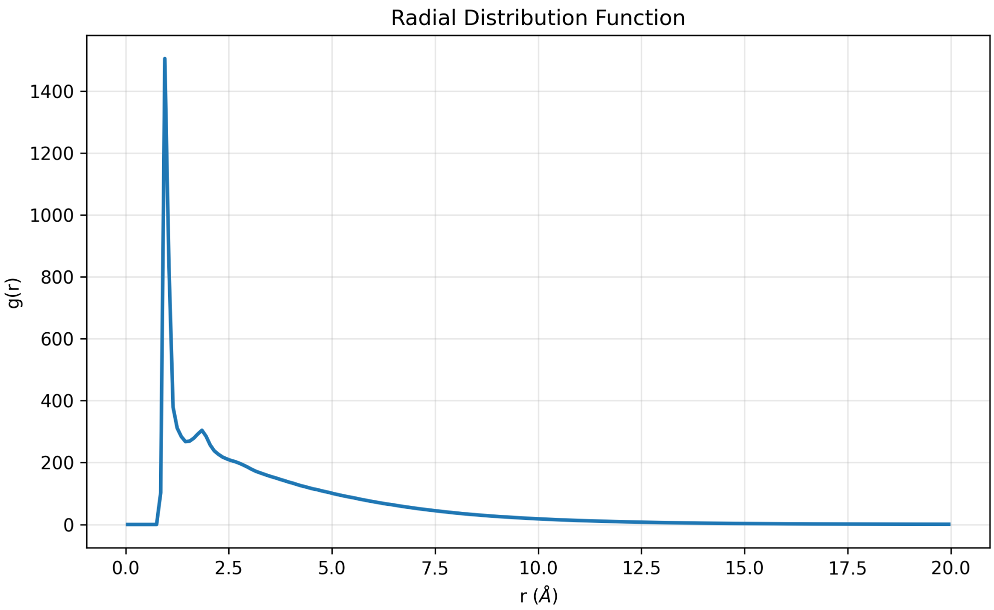

Figure 2.3: Isotropic Radial Distribution Function $g(r)$ showing local coordination shells.

Chapter 3

Protein–DNA Target Search Analysis

3.1 Introduction and Objectives

DNA-binding proteins locate target operator sites remarkably quickly. This process is modeled using ”facilitated diffusion,” where the protein switches between 3D bulk diffusion through the solution and 1D sliding or crawling along the DNA strand. The goal of this analysis was to classify these search modes from coordinate data and see how changing the attractive interaction strength affects the timing.

3.2 Simulation Specifics and Setup

The Python scripts read 3D coordinate trajectories tracking two components: a 101-monomer DNA chain and a single mobile protein particle. I looked at four separate interaction energy well depths: $\epsilon = 3, 4, 5$, and 6. Trajectory frames were saved every 1000 simulation steps. With an integration step size of $dt = 0.005$, the time separating consecutive frames is exactly $1000 \times 0.005 = 5$ time units. To get reliable averages, I processed 300 independent simulation runs for each ϵ setting.

The interaction parameter ϵ adjusts how strongly the protein is attracted to the DNA. A higher ϵ means the protein wants to stay bound to the DNA surface longer, changing the balance between free 3D diffusion and localized 1D sliding.

3.3 Methodology

3.3.1 Distance Calculations and Periodic Edges

For every frame, I calculated the distance between the protein and each monomer $jontheDNAchain.Toh$ $x_{\mathrm{DNA},j} - x_{\mathrm{protein}} - L_x \cdot \mathrm{round}\left(\frac{x_{\mathrm{DNA},j} - x_{\mathrm{protein}}}{L_x}\right)$ (3.1) This calculation was mirrored for the y and z axes. The absolute distance to each monomer was calculated using:

$$d_j = \sqrt{(\Delta x_j)^2 + (\Delta y_j)^2 + (\Delta z_j)^2} \tag{3.2}$$

The immediate position of the protein relative to the DNA was tracked using the global minimum distance in that frame:

$$d_{\min} = \min_{j \in [1,101]} \{d_j\} \tag{3.3}$$

3.3.2 Search Mode Classification

I used a spatial cutoff distance of $d_{\mathrm{cutoff}} = 2.5\sigma$ to determine the protein's state:

- **1D Mode:** If $d_{\min} < 2.5\sigma$, the protein is close enough to be inside the DNA interaction cylinder (sliding or hopping).
- **3D Mode:** If $d_{\min} \geq 2.5\sigma$, the protein is out in the bulk volume (diffusing freely).

The total time spent in each mode was calculated by multiplying the frame counts by the time step size:

$$\tau_{1D} = N_{1D} \times 5, \quad \tau_{3D} = N_{3D} \times 5 \tag{3.4}$$

The script also counted every time the protein crossed the 2.5σ boundary line, tracking the number of mode switches (N_s).

3.4 Results and Kinetic Analysis

The averages across all 300 runs are summarized in Table 3.1.

Interaction Strength (ϵ)	Mean 1D Time ($\langle\tau_{1D}\rangle$)	Mean 3D Time ($\langle\tau_{3D}\rangle$)	Mean Switche
3	712.37	8613.70	15.94
4	1886.18	4794.38	8.65
5	4913.03	2621.13	4.20
6	8359.05	1713.82	1.90

Table 3.1: Protein–DNA facilitated diffusion metrics across varying interaction strengths.

The metrics reveal a strong dependency on the binding affinity. When ϵ is low ($\epsilon = 3$), the protein stays mostly in the bulk space ($\langle \tau_{3D} \rangle = 8613.70$) because thermal energy ($k_B T$) easily breaks the weak attraction. As the attraction scales up to $\epsilon = 6$, the average 1D sliding time $\langle \tau_{1D} \rangle$ jumps significantly to 8359.05, while the 3D exploration time drops to 1713.82.

At the same time, the average number of switches $\langle N_s \rangle$ falls from 15.94 down to 1.90. This indicates that a deep potential energy well creates a high desorption barrier; once the protein lands on the DNA, it struggles to escape back into the bulk volume. Figure 3.1 shows this clear trade-off between 1D surface crawling and 3D volume sampling as the interaction strength changes.

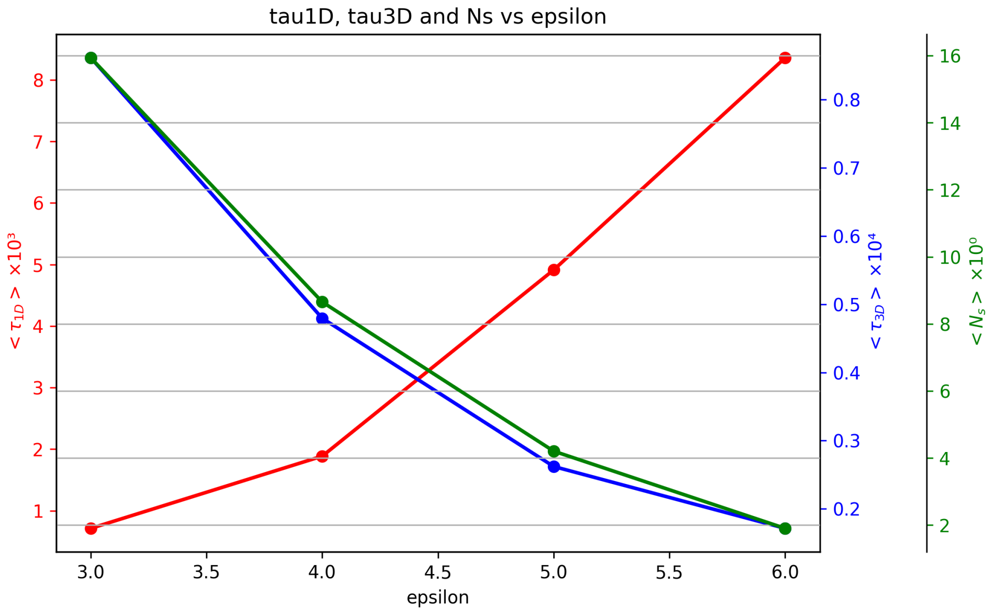

Figure 3.1: Average 1D sliding time, 3D bulk time, and total switches plotted against interaction strength ϵ.

Chapter 4

Polymer Dynamics During Annealing

4.1 Introduction and Timestep Segmentation

To study non-equilibrium dynamics and structural relaxation, I looked at a polymer system undergoing a controlled cooling cycle. The coordinates were recorded in unwrapped formats to maintain continuous distance paths. The trajectory data was split into seven separate time blocks representing sequential cooling stages, as listed in Table 4.1.

Annealing Block Index	**Timestep Boundaries Range**
1	0.0×10^7 to 2.5×10^7
2	2.5×10^7 to 5.0×10^7
3	5.0×10^7 to 7.5×10^7
4	7.5×10^7 to 1.0×10^8
5	1.0×10^8 to 1.25×10^8
6	1.25×10^8 to 1.5×10^8
7	1.5×10^8 to 1.75×10^8

Table 4.1: Timestep segmentation mapping across the seven thermal annealing blocks.

4.2 Methodology

4.2.1 Mean Squared Displacement (MSD)

Monomer mobility within each temperature block was measured using the Mean Squared Displacement:

$$\mathrm{MSD}(\tau) = \langle |\mathbf{r}(t+\tau) - \mathbf{r}(t)|^2 \rangle \tag{4.1}$$

where $\mathbf{r}(t)$ is the unwrapped position at time t, τ is the lag time, and the average is computed over all time origins t and all monomers. I used an FFT-based autocorrelation script to calculate this quickly over long trajectories.

4.2.2 Power-law Scaling Regression

To characterize the mode of transport diffusion, the stable early-time regimes of the MSD curves were fitted to a power-law relationship:

$$\mathrm{MSD}(\tau) = A\tau^{\alpha} \tag{4.2}$$

where A is a diffusion prefactor and α is the scaling exponent. An exponent of $\alpha = 1$ indicates standard Brownian diffusion, while $\alpha < 1$ shows subdiffusive behavior. To avoid statistical noise at long lag times where data points get sparse, the fitting window was strictly limited to the first half of each block's duration ($T \leq 1/2$).

4.3 Results and Discussion

4.3.1 Anomalous Transport Dynamics

The power-law fitting outputs across the seven blocks are shown in Table 4.2.

Block	**Prefactor Amplitude (A)**	**Diffusion Exponent (α)**
1	1.0149×10^{0}	0.3591
2	8.2868×10^{-1}	0.4107
3	6.7662×10^{-1}	0.5551
4	7.5000×10^{-1}	0.6015
5	4.5108×10^{-1}	0.4067
6	4.2879×10^{-1}	0.4450
7	3.0072×10^{-1}	0.4942

Table 4.2: Power-law fitting parameters extracted from blockwise MSD analysis.

All blocks show diffusion scaling exponents well below 1.0 ($\alpha \approx 0.36 - 0.60$), confirming subdiffusive motion. This slow movement is typical for polymer chains experiencing entropic traps and steric crowding from the matrix particles. The exponent rises slightly during the intermediate blocks (Blocks 2–4), peaking at 0.6015, which suggests a temporary window of higher relative mobility before dropping back down as the system cools further.

Figure 4.1 shows the MSD curves plotted on log-log axes. The curves shift downward vertically across consecutive blocks, mapping the drop in temperature as kinetic energy is removed. At very long lag times ($\tau > 10^4$), the lines begin to scatter and show statistical noise. This occurs because fewer overlapping time frames are available to average at the end of a block, justifying why the fitting window was limited to the first half of the block duration.

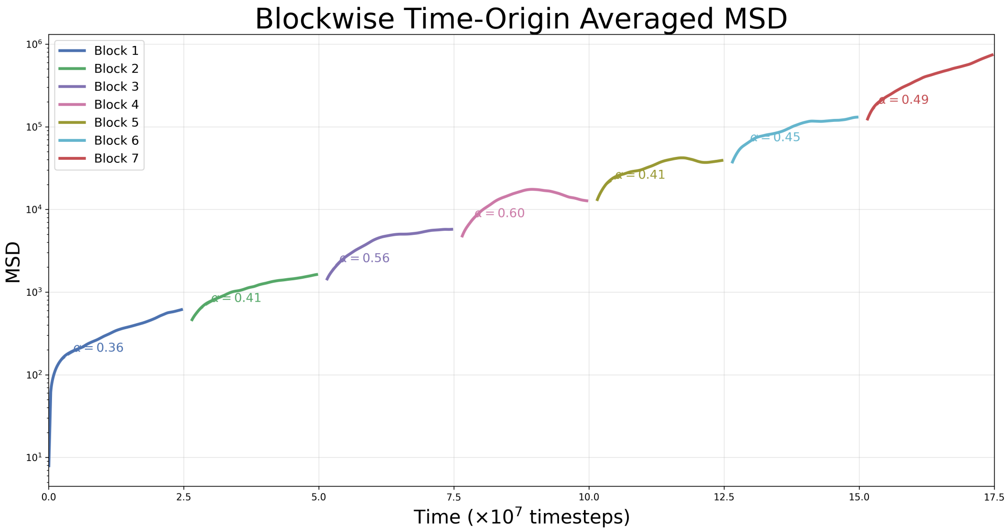

Figure 4.1: Log-log plot of Mean Squared Displacement (MSD) vs lag time across the seven temperature blocks.

4.3.2 Blockwise Radius of Gyration

The change in physical chain dimensions across the annealing blocks is tracked in Table 4.3.

Block	**Radius of Gyration ($R_g \pm$ SD)**
1	15.2032 ± 1.5248
2	12.8645 ± 1.4233
3	11.3212 ± 1.4006
4	8.7162 ± 0.6558
5	7.9530 ± 0.1581
6	7.4845 ± 0.3029
7	7.0173 ± 0.0596

Table 4.3: Mean blockwise Radius of Gyration values showing progressive compaction.

The mean R_g decreases steadily from 15.2032 in Block 1 down to 7.0173 in Block 7. This step-by-step reduction is plotted in Figure 4.2 and confirms that the polymer is compressing uniformly as it cools. The standard deviations also shrink in the later blocks, which makes physical sense because cooling drains thermal energy, locking the chain into a stable compact structure and dampening large size fluctuations.

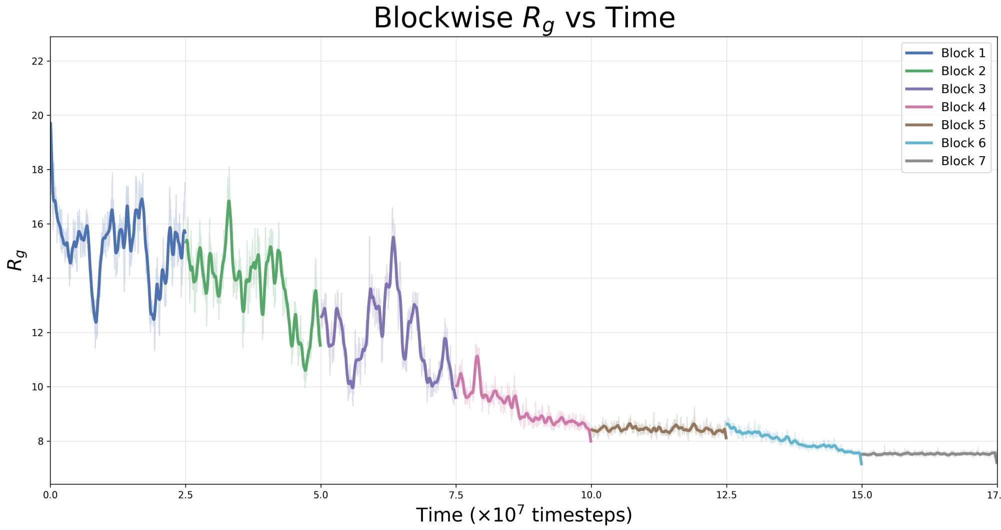

Figure 4.2: Average Radius of Gyration for each cooling block, showing the chain compressing.

Chapter 5

Static Structure Factor Analysis

5.1 Introduction and Formalism

To map the spatial arrangement of the polymer chain across different length scales, I calculated the static structure factor, $S(\mathbf{k})$. This reciprocal-space metric is useful because it links the coordinate history directly to scattering profiles from experimental techniques like SAXS or SANS.

The analysis focused on the main chain (**atom type 1**) inside the trajectory file `Nb100Ns20gd1.0.lmptrj`. The box dimensions were $L_x = 60, L_y = 40$, and $L_z = 150$. I sampled 20 evenly spaced coordinate snapshots across the first four large annealing blocks (from step 0 to 1.0×10^8).

The static structure factor was computed using spatial vector sums:

$$S(\mathbf{k}) = \frac{1}{N} \left\langle \left| \sum_{j=1}^{N} e^{i\mathbf{k}\cdot\mathbf{r}_j} \right|^2 \right\rangle \tag{5.1}$$

where $N = 100$ is the number of backbone monomers and $\mathbf{r}_j$ represents their coordinates. Valid wave vectors ($\mathbf{k}$) matching the periodic bounds were generated using integer steps:

$$k_x = \frac{2\pi n_x}{L_x}, \quad k_y = \frac{2\pi n_y}{L_y}, \quad k_z = \frac{2\pi n_z}{L_z} \tag{5.2}$$

To turn this into a 1D profile, I used a shell-averaging routine that groups wave vectors by their scalar magnitude ($k = |\mathbf{k}|$) into thin radial bins and takes the average intensity.

5.2 Results and Structural Interpretation

The structural properties across length scales were analyzed by fitting the linear regions on a log-log plot to a power law ($S(k) \propto k^{-\alpha}$). The exponents for Blocks 3 and 4 are listed in Table 5.1.

Annealing Block	Wave Vector Range (k)	Extracted Exponent (α)
Block 3	$0.10 < k < 0.30$	1.564
Block 3	$0.80 < k < 2.50$	1.410
Block 4	$0.10 < k < 0.30$	1.478
Block 4	$0.80 < k < 2.50$	1.335

Table 5.1: Power-law exponents extracted from blockwise static structure factor curves.

The scattering profiles shown in Figure 5.1 highlight clear physical regimes. At low wave vectors ($k < 0.3$), which probe large spatial dimensions, we see an increase in $S(k)$. This upward trend matches the physical picture from our real-space R_g data, tracking the large-scale contraction of the chain mass. The extracted exponents ($\alpha \approx 1.3 - 1.6$) fall within the expected range for compacted polymer geometries.

At high wave vectors ($k > 2.5$), which look at tiny, local length scales, the scattering curves flatten out into a baseline floor near $S(k) \approx 1$. At this short distance scale, the calculation is probing lengths smaller than the monomer separations, where independent, un-correlated scattering takes over. Block 4 displays a noticeable non-linear crossover region around $k \approx 0.5$. This represents the transition boundary where the structural geometry changes from large global dimensions down to localized sub-chain configurations.

Constraint at Low k: My mentor suggested checking the power-law scaling at very low wave vectors ($k < 0.1$). However, when testing the script, I found that the minimum available wave vector magnitude was bounded at $k \approx 0.128$ due to the finite size of the simulation box ($L_y = 40$). This meant no data points existed below 0.1, preventing an independent power-law fit in that region.

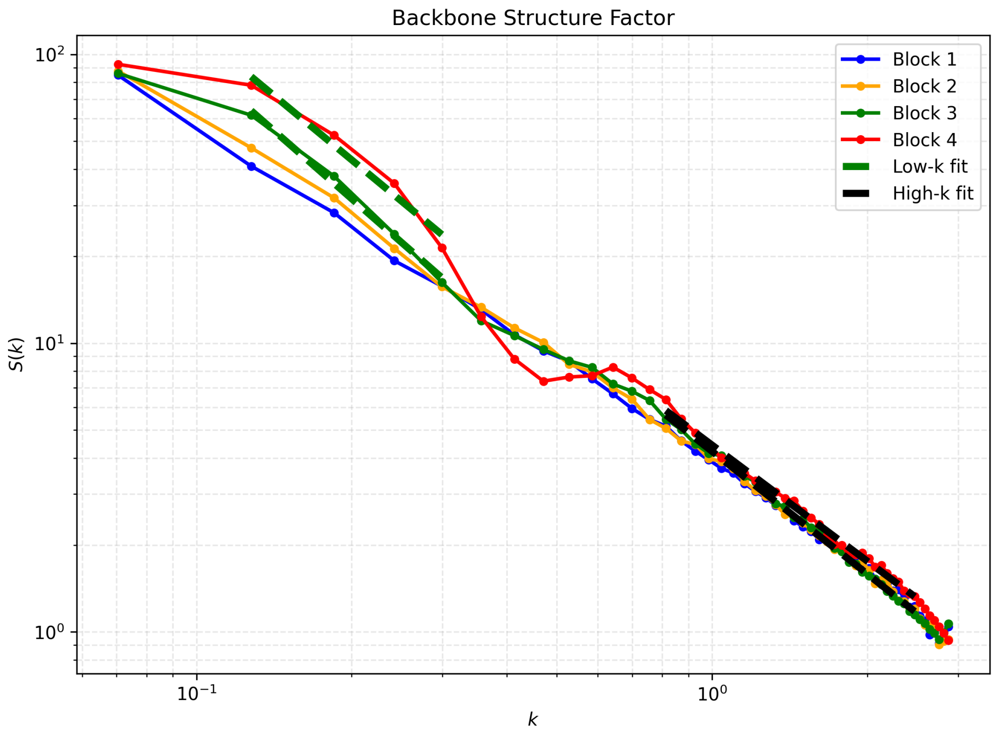

Figure 5.1: Log-log plot of the static structure factor $S(k)$ showing power-law fits for different blocks.

Chapter 6

Computational Optimizations

Processing MD trajectories containing millions of coordinate rows can be very slow. To keep the execution times manageable on my system, I focused on a few core computational optimizations:

- **NumPy Vectorization:** I eliminated nested Python loops in the coordinate processing steps. Distance matrix calculations and minimum-image periodic boundary checks were rewritten using NumPy broadcasting, which handles the calculations in optimized, compiled C code.
- **FFT-Based Correlation:** A standard Mean Squared Displacement calculation scales quadratically as $\mathcal{O}(M^2)$ with the number of frames. By using a Fast Fourier Transform (FFT) approach based on the Wiener-Khinchin theorem, I reduced the scaling to $\mathcal{O}(M \log M)$, which drastically speeded up the lag-time calculations.
- **Memory Handling:** To stop the scripts from running out of RAM, I used contiguous memory allocations for distance matrices and recycled arrays across frames. Switching to `float32` data precision where appropriate cut memory usage during massive matrix steps.

Script Validation Note: To ensure these optimized pipelines weren't introducing mathematical bugs, I ran sanity checks using small, synthetic coordinate text files. I hand-calculated the R_g and pairwise distances for a simple 5-particle system across 10 frames and verified that the optimized Python parser reproduced the exact analytical answers perfectly.

These changes dropped computational costs significantly, allowing the analysis of trajectories containing thousands of frames and millions of coordinate entries.

Chapter 7

Integrated Discussion and Conclusion

7.1 Overall Discussion

Looking at the data from both projects together gives us a clear look at how these macromolecular systems behave under different conditions.

In the polymer project, both the equilibrium measurements and the thermal annealing data track a continuous structural shrinking. The chain transitions from a relatively expanded shape down to a highly compact state. This contraction is mapped out clearly by the falling Radius of Gyration (R_g) values, and it is supported by the tight short-range coordination shells seen in the RDF peaks. The subdiffusive scaling exponents calculated from the MSD curves ($\alpha < 1$) show that monomer motion is restricted by the surrounding crowding matrix. The scattering shapes from the static structure factor ($S(k)$) echo this structural change, with scaling exponents ($\alpha \approx 1.3 - 1.6$) typical of a highly compressed conformation.

At the same time, the target search project shows how closely a protein's movement is tied to its attraction affinity (ϵ) to DNA. Increasing the binding affinity locks the protein into the 1D search space, which boosts $\langle \tau_{1D} \rangle$ and suppresses the switching frequency $\langle N_s \rangle$. This helps us quantify the physical speed limits and trade-offs involved in facilitated diffusion mechanisms.

7.2 Summary of Key Results

Table 7.1 groups the core numerical metrics collected across the different analysis pipelines.

Physical Evaluated Metric	**Numerical Result / Extracted Window**
Persistence Length (l_p)	$0.98\,\sigma$
Initial Mean Radius of Gyration (R_g Initial)	$15.20\,\sigma$
Final Compacted Radius of Gyration (R_g Final)	$7.01\,\sigma$
MSD Anomalous Scaling Range (α)	0.36 – 0.60
Static Structure Factor $S(k)$ Scaling Exponents	1.33 – 1.56

Table 7.1: Consolidated summary of structural, transport, and scattering results.

In conclusion, these optimized analysis pipelines successfully turn raw LAMMPS coordinates into clear physical parameters. The automated tools I wrote are flexible enough to be reused for more complex setups, like multi-protein environments or different thermal cooling profiles. This internship gave me great hands-on experience in handling large MD datasets, writing optimized data code, and connecting numerical results to physical scaling theories.

Future Work

The analyses carried out during this internship provide a foundation for several possible extensions that could further improve the understanding of protein–DNA target search and polymer dynamics.

- **Longer Simulation Times:** Extending the duration of the molecular dynamics simulations would improve statistical sampling, particularly at large lag times. This would reduce fluctuations in the Mean Squared Displacement (MSD) and static structure factor, resulting in more reliable estimates of long-time dynamical behavior.
- **Extended Parameter Exploration:** Investigating additional values of the interaction strength (ϵ) and employing slower annealing schedules would provide a more detailed understanding of the transition between different structural and dynamical regimes. Such studies could help establish a comprehensive phase diagram describing the polymer's response to varying interaction strengths and thermal conditions.
- **Multi-Protein Target Search:** The current analysis considers the dynamics of a single diffusing protein. Extending the methodology to systems containing multiple proteins would enable the investigation of cooperative effects, molecular crowding, and competitive binding, thereby providing a more realistic representation of intracellular target-search processes.

References

1. Plimpton, S., “Fast Parallel Algorithms for Short-Range Molecular Dynamics,” *Journal of Computational Physics*, 117(1), 1–19, 1995.

2. Thompson, A. P. *et al.*, “LAMMPS – A Flexible Simulation Tool for Particle-Based Materials Modeling at the Atomic, Mesoscopic, and Continuum Scales,” *Computer Physics Communications*, 271, 108171, 2022.

3. Doi, M. and Edwards, S. F., *The Theory of Polymer Dynamics*, Oxford University Press, 1986.

4. de Gennes, P. G., *Scaling Concepts in Polymer Physics*, Cornell University Press, 1979.

5. Rubinstein, M. and Colby, R. H., *Polymer Physics*, Oxford University Press, 2003.

6. Berg, O. G., Winter, R. B., and von Hippel, P. H., “Diffusion-Driven Mechanisms of Protein Translocation on Nucleic Acids,” *Biochemistry*, 20(24), 6929–6948, 1981.

7. Halford, S. E. and Marko, J. F., “How Do Site-Specific DNA-Binding Proteins Find Their Targets?” *Nucleic Acids Research*, 32(10), 3040–3052, 2004.

8. Metzler, R. and Klafter, J., “The Random Walk’s Guide to Anomalous Diffusion,” *Physics Reports*, 339(1), 1–77, 2000.

9. Hansen, J. P. and McDonald, I. R., *Theory of Simple Liquids*, 4th Edition, Academic Press, 2013.

10. Frenkel, D. and Smit, B., *Understanding Molecular Simulation: From Algorithms to Applications*, 2nd Edition, Academic Press, 2002.